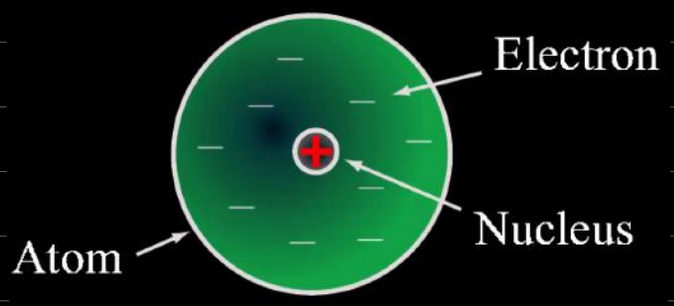
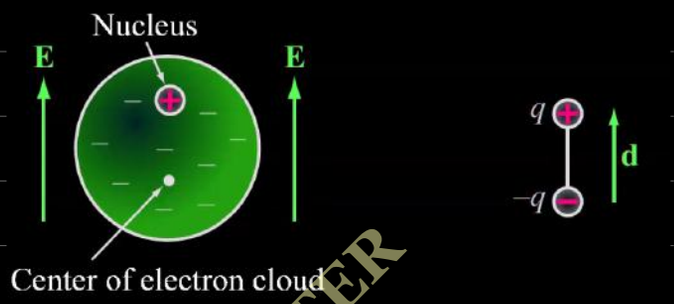
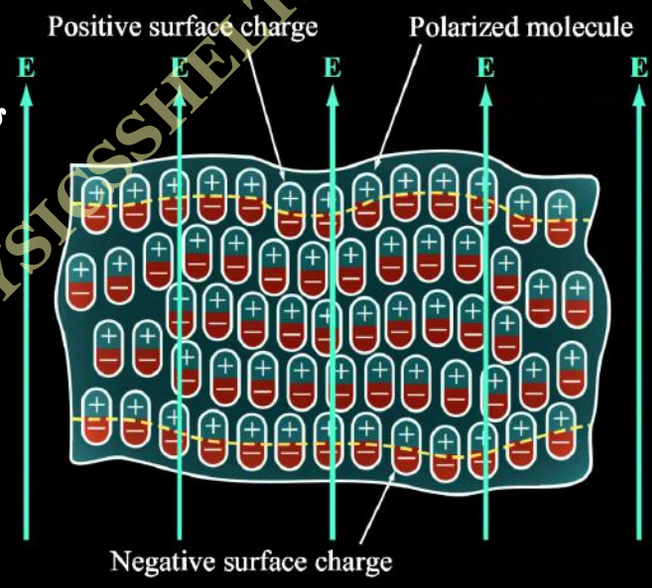
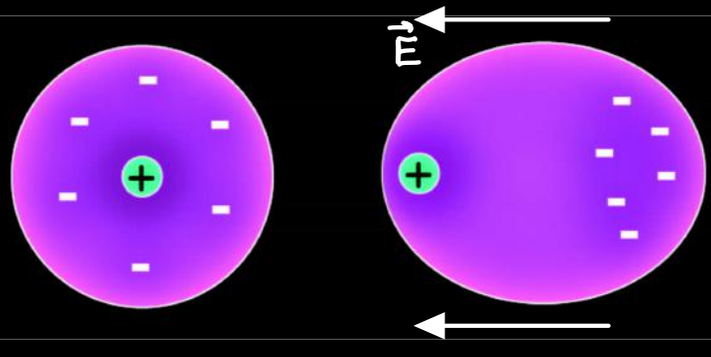
POLARIZATION: Consider a non polar atom, under normal circumstances, we expect average location of positive charge (nucleus) to coincide with average location of negative charge (electron cloud), thus producing a net zero dipole moment. However when such a molecule is subjected to external field, the field pushes the nucleus and pulls the electron cloud, thus producing a net dipole moment of qd where d is the relative separation between positive and negative charge centers. In a homogenious dielectric its net effect is equivalent to creating surface positive and negative charge densities while the matrix being still neutral (At a macroscopic level). While for a non homogeneous dielectric there may also be net charge in the volume.







POLARIZABILITY: For a typical molecule, the induced dipole moment is proportional to the applied electric field, polarizability is a measure of this dipole moment produced per unit applied electric field. Quantitatively:



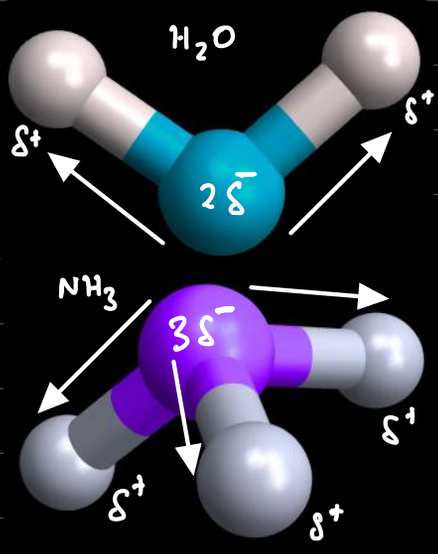
$$\vec{p} = \beta \epsilon_0 \vec{E}$$

Here β is polarizability, $\vec{E}$ local macroscopic electric field and $\vec{p}$ is dipole moment of molecule.

(This is as per definition of russian books, Purcell gives $\vec{p} = \alpha \vec{E}$ and with polarizability being ambiguously either α or $\alpha/4\pi\epsilon_0$, in any case all definitions only differ by a multiplied constant only).

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Polar Dielectrics: Some molecules have permanent dipole moments due to different electron affinities of elements and molecular structures (such as H_2O or NH_3). Such molecules experience a torque when exposed to an electric field and tend to orient their net dipole moments along the applied electric field. The degree of alignment depends on the strength of applied electric field. When no field is applied, net dipole moment is zero. As the strength of applied field is increased, the effect of random thermal oscillations is dominated by aligning effect of the applied field. Macroscopic effect of applying field to polar dielectrics is similar to that of non polar dielectrics (causing effectively a net surface charge density), however here the polarisation is much more stronger. Here also we can talk of polarisability as a measure of net average dipole moment produced per molecule for an applied field. For polar molecules, polarizability reduces with temperature due to increased random thermal oscillations.



Polarization Vector: Net dipole moment of a substance per unit volume is called polarization vector. If there are n dipoles per unit volume, then net dipole moment vector is obtained as:

$$\vec{p}_{net} = \sum_{i=1}^{n\Delta V} \vec{p}_i \quad \text{for a volume } \Delta V$$

thus local polarization vector at any point in a dielectric is defined as:

$$\vec{P} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \sum_{i=1}^{n\Delta V} \vec{p}_i = \frac{\vec{p}_{net}}{\Delta V}$$

(Here we treat $\vec{P}$ at macroscopic level as a continuous quantity though at microscopic level (e.g. inside an atom) it may not be defined. ΔV may be very small but still large enough to accomodate

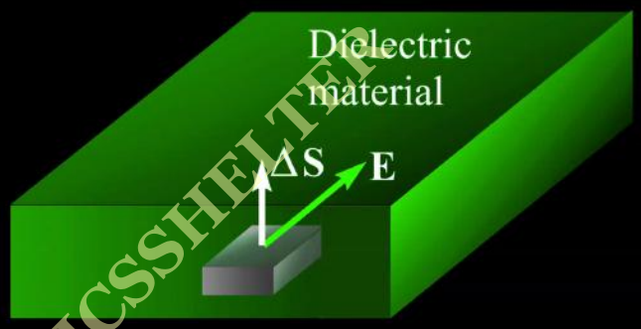
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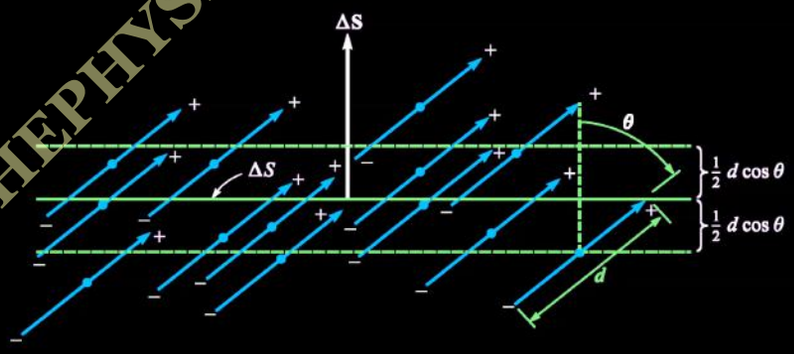
sufficient number of dipoles for a statistical averaging).

Bound Charge: Any charge not capable of large macroscopic movements due to being paired with a charge of opposite polarity (such as all dielectric materials) is called bound charge. For a dielectric if there is net such charge in a volume, it is called volume bound charge while if it is on surface, we call it surface bound charge.

Bound Charge Crossing A Surface:

We consider a small patch of area ΔS inside a dielectric substance as shown in a non polar dielectric. Now when an electric field is applied some positive charge crosses this S upwards and negative charge crosses downwards. If d is the separation between positive and negative charges, $p = q d$ is dipole moment of each molecule. Now any positive charge lying within a distance $\frac{1}{2} d \cos\theta$ must have crossed ΔS upwards





similarly any charge lying within a distance $\frac{d}{2}\cos\theta$ must have crossed it downwards. Now if n is the number of charges per unit volume, positive charge crossing upwards, $\Delta q_b^+ = \Delta S \times \frac{d}{2}\cos\theta \times n q$

similarly negative charge crossing downwards, $\Delta q_b^- = \Delta S \frac{d}{2}\cos\theta \times n q$

Thus net algebraic charge crossing outwards (towards area normal)

$$\Delta q_b = \Delta q_b^+ + \Delta q_b^- = \Delta S\, n q\, d \cos\theta \quad (1)$$

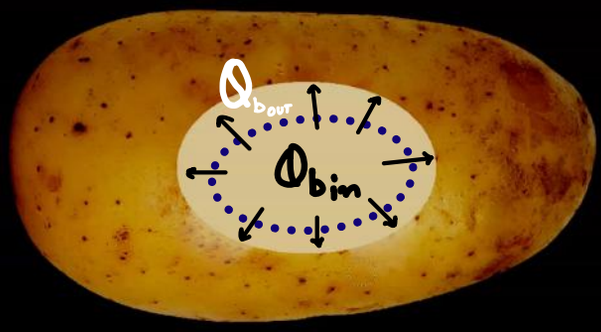
Now polarization vector : $\vec{P} = n q d \hat{E}$ (2) (∵ n is dipoles/unit volume).

From (1) & (2):

$$\boxed{\Delta q_b = \vec{P} \cdot \Delta \vec{S}} \quad \text{or} \quad dq_b = \vec{P} \cdot d\vec{S} \quad (1)$$

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Now this equation holds for every small patch inside dielectric, to find the net charge flown out of a closed surface we can integrate the equation ① over entire closed surface and use the fact that since overall dielectric was neutral, thus charge flown out of closed surface must be negative of charge enclosed inside the surface. Thus:



$$\oint \vec{P} \cdot d\vec{S} = Q_{b_{out}} = -Q_{b_{in}}$$

Thus: $$\boxed{-\oint \vec{P} \cdot d\vec{S} = Q_{b_{in}}}$$

SUSCEPTIBILITY: Whenever there is a net electric field $\vec{E}$ inside a dielectric material, the polarization is directly proportional to the $\vec{E}$. Susceptibility χ is a measure of polarisation produced per unit electric field and is a material property. It is defined by following equation:

$$\boxed{\vec{P} = \chi \epsilon_0 \vec{E}}$$

→ Polar dielectrics have higher susceptibility
→ Susceptibility of polar molecules decreases with temperature.

DIELECTRIC CONSTANT: In many expressions in study of dielectrics we come across the factor of $1+\chi$ and we give it a term "Dielectric constant K" thus:

$$\boxed{K = 1 + \chi}$$

GAUSS LAW FOR DIELECTRICS:

Recall that $$-\oint \vec{P} \cdot d\vec{S} = Q_{b_{in}} = -\oint \chi \epsilon_0 \vec{E} \cdot d\vec{S} \quad ①$$

also by usual Gauss law, $$\oint \vec{E} \cdot d\vec{S} = Q_{in} = (Q_{free} + Q_{b_{in}})/\epsilon_0 \quad ②$$

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Here Q_{free} refers to any extraneous charges that we may have embedded inside the dielectric (Like charged metallic particles). Now rearranging equations ① and ② we get:

$$\oint \chi \epsilon_0 \vec{E} \cdot d\vec{S} = -Q_{bin} \quad ③$$

and $$\oint \epsilon_0 \vec{E} \cdot d\vec{S} = Q_{free} + Q_{bin} \quad ④$$, Now adding we get:

$$\oint \epsilon_0 (1+\chi) \vec{E} \cdot d\vec{S} = Q_{free}$$

$$\Rightarrow \boxed{\oint K\vec{E} \cdot d\vec{S} = \frac{Q_{free}}{\epsilon_0}} \leftarrow \text{GAUSS LAW FOR DIELECTRICS.}$$

→ For homogeneous dielectrics, K can be taken outside the integral

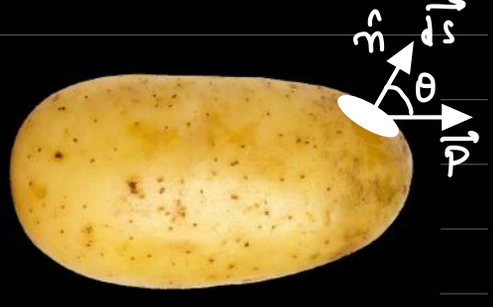
→ If Gaussian surface passes through metal matrix, $\int K\vec{E} \cdot d\vec{S}$ must be taken as zero for that part.

EXAMPLE: Prove that a homogeneous dielectric cannot have any volume bound charge density.

Solution: $\oint K\vec{E} \cdot d\vec{S} = 0$, multiplying both sides by $\chi\epsilon_0/K$,

$$\oint \chi\epsilon_0 \vec{E} \cdot d\vec{S} = 0 \Rightarrow \oint \vec{P} \cdot d\vec{S} = 0 \Rightarrow Q_{bin} = 0$$, thus a homogeneous dielectric cannot even have a volume bound charge density, It can only have surface bound charge density.

SURFACE BOUND CHARGE DENSITY IN TERMS OF LOCAL POLARIZATION:



Recall that charge crossing any surface is given as $dq = \vec{P} \cdot d\vec{S}$, now consider a patch at the surface of dielectric, whatever charge crosses a patch very close to surface must reside at the surface, Now $\frac{dq}{dS} = \frac{\vec{P} \cdot d\vec{S}}{dS}$

$$\Rightarrow \boxed{\sigma_b = \vec{P} \cdot \hat{n}}$$

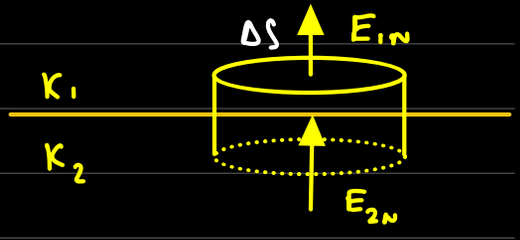
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PERMITTIVITY OF A MEDIUM: The product $\epsilon_0 K$ is called permittivity of any medium, mathematically, $\epsilon = \epsilon_0 K$. (Sometime K is called relative permittivity, or $\epsilon = \epsilon_0 \epsilon_r$).

→ By permittivity I mean always means relative permittivity (dielectric constant).

BOUNDARY CONDITION FOR NORMAL FIELD AT DIELECTRIC BOUNDARY:

Consider a pill shaped Gaussian surface near the boundary of two dielectric with dielectric constants K_1 and K_2. There is no free charge enclosed at the boundary. Now applying Gauss law for dielectrics:



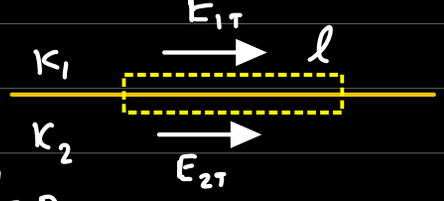
$$\oint K\vec{E}\cdot d\vec{S} = \frac{Q_{free}}{\epsilon_0} \Rightarrow K_1 E_{1N} S - K_2 E_{2N} S = 0$$

(Here E_{1N} & E_{2N} are field components ⊥ to the boundary), thus:

$$\boxed{K_1 E_{1N} = K_2 E_{2N}}$$

BOUNDARY CONDITION FOR TANGENTIAL FIELDS:

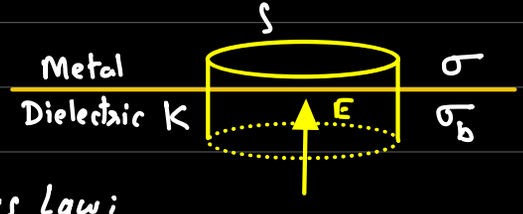
Consider closed loop near the boundary of two dielectrics. Net work done along this closed path must be zero. Thus, $E_{1T} \times l - E_{2T} \times l = 0$



$$\Rightarrow \boxed{E_{1T} = E_{2T}}$$

BOUND CHARGE DENSITY AT METAL DIELECTRIC BOUNDARY:

Let σ be the free charge density on metal surface and σ_b be the bound charge density on dielectric surface. Now consider the pill shaped Gaussian surface as shown. Now by Gauss Law:



$$\oint \vec{E}\cdot d\vec{S} = \frac{Q_{in}}{\epsilon_0} \Rightarrow -ES + 0 = (\sigma + \sigma_b) S/\epsilon_0 \quad (1)$$

Now applying Gauss law for Dielectrics $\oint K\vec{E}\cdot d\vec{S} = -KES = \sigma S/\epsilon_0$ (2)

Now substituting $-ES$ from (1) in (2):

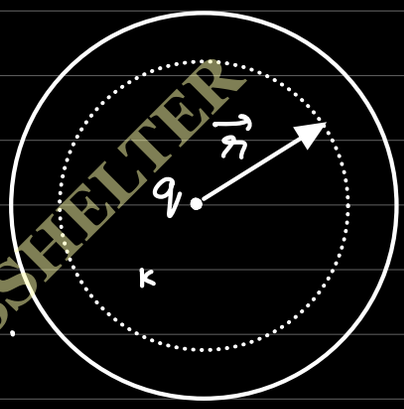
$$\frac{K(\sigma_b + \sigma)S}{\epsilon_0} = \frac{\sigma S}{\epsilon_0}$$

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$$\Rightarrow \quad \boxed{\sigma_b = -\sigma\left(1 - \frac{1}{K}\right)} \leftarrow \text{Always holds at metal dielectric boundary.}$$

Example: Irodov 3.74

A point charge q is located at the centre of a ball made of uniform isotropic dielectric with constant k. Find the polarization **P** as a function of the radius vector $\vec{\mathbf{r}}$ relative to the centre of the system, as well as the charge q_b inside a sphere whose radius is less than the radius of the ball.



Solution: Imagine a gaussian sphere of radius r.

By Gauss Law for Dielectrics,

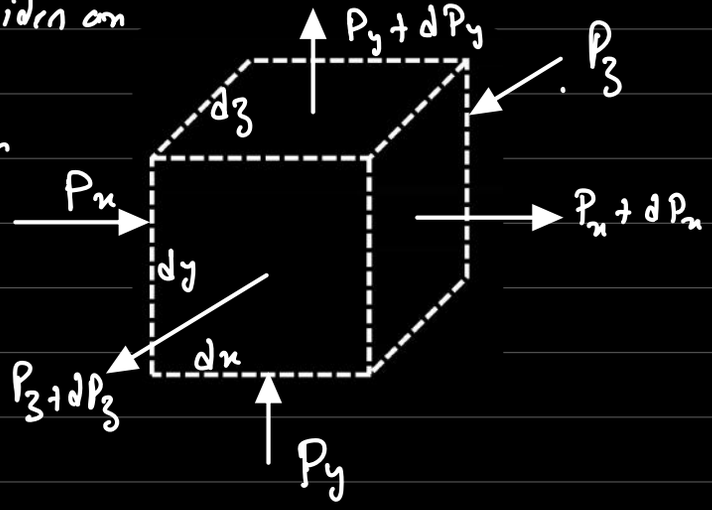
$$\chi E \times 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi\epsilon_0 K}\frac{q}{r^2}$$

Now $\vec{P} = \chi \epsilon_0 \vec{E} \Rightarrow \vec{P} = \left(\frac{K-1}{K}\right)\frac{1}{4\pi}\frac{q}{r^3}\vec{r}$ Ans.

$$q_b = -P \times 4\pi R^2 = -(K-1/K)\,q$$

Differential Relation Between Polarization Vector And Bound Charge Density:

Recall that $\oint \vec{P}\cdot d\vec{S} = -Q_b$, now consider an elemental cube, applying the equation to this cubical surface:



$$(P_x + dP_x)dy\,dz - P_x dy dz$$
$$+(P_y + dP_y)dx\,dz - P_y\,dx\,dz$$
$$+(P_z + dP_z)dx\,dy - P_z\,dx\,dy = -\rho_b\,dx\,dy\,dz$$

Rearranging we get:

$$\frac{dP_x}{dx} + \frac{dP_y}{dy} + \frac{dP_z}{dz} = -\rho_b$$

(Here actually we mean partial derivatives since P is a multivariate function). Thus:

$$\frac{\partial P}{\partial x} + \frac{\partial P}{\partial y} + \frac{\partial P}{\partial z} = -\rho_b \quad \text{or} \quad \boxed{\vec{\nabla}\cdot\vec{P} = -\rho_b}$$

This means for uniformly polarised substance, there is no volume

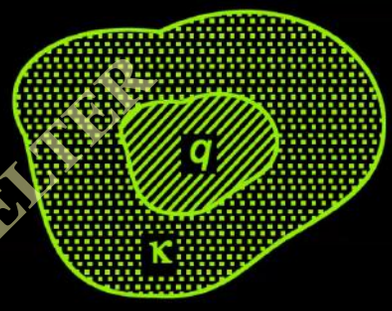
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bound charge density.

STATIC POLARIZATION: In certain substances a permanent dipole moment can be created using external electric field and heating and cooling the substances. Such substances are said to be statically charged.

EXAMPLE IRODOV 3.76:

A conductor of arbitrary shape, carrying a charge q, is surrounded with uniform dielectric of constant κ.
Find the total bound charges at the inner and outer surfaces of the dielectric.



Solution: Recall that charge density at metal dielectric boundary is given as: $\sigma_b = -\sigma(1 - 1/\kappa)$, also for conductor, the entire charge resides on surface, thus total bound charge on inner boundary of dielectric is given as:

$$Q_b = \oint \sigma_b \, dS = -\left(1 - \frac{1}{\kappa}\right) \int \sigma \, dS = -q\left(1 - \frac{1}{\kappa}\right)$$ Ans.

Now since there are no volume bound charges (∵ homogeneous dielectric) thus outer surface of dielectric must get exactly negative q charge at inner boundary i.e $q(1 - 1/\kappa)$ Ans.

EXAMPLE: IRODOV 3.78:

Near the point A (Fig. 3.10) lying on the boundary between glass and vacuum the electric field strength in vacuum is equal to $E_0 = 10.0$ V/m, the angle between the vector $\mathbf{E}_0$ and the normal $\mathbf{n}$ of the boundary line being equal to $\alpha_0 = 30°$. Find the field strength E in glass near the point A, the angle α between the vector $\mathbf{E}$ and $\mathbf{n}$, as well as the surface density of the bound charges at the point A. (κ is given)

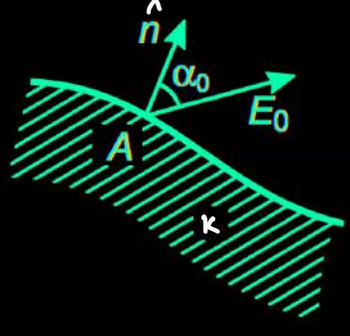


Fig. 3.10

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Solution: $K_1 E_{1N} = K_2 E_{2N}$ $\Rightarrow$ Normal field in dielectric $= E_N = \frac{E_0 \cos\alpha_0}{K}$

By continuity of tangential field, $E_T = E_0 \sin\alpha_0$.

now let angle of field lines with $\hat{n}$ inside be α, then, $\tan\alpha = E_T/E_N$

$\Rightarrow$ $\alpha = \tan^{-1}(K \tan\alpha_0)$ Ans.

Now bound charge density $\sigma_b = \vec{P}\cdot\hat{n} = P_N = \chi \epsilon_0 E_N$

$\Rightarrow$ $\sigma_b = (K-1)\epsilon_0 \frac{E_0 \cos\alpha_0}{K} = \left(\frac{K-1}{K}\right)\epsilon_0 E_0 \cos\alpha_0$ Ans.

EXAMPLE : IRODOV : 3.79:

Near the plane surface of a uniform isotropic dielectric with constant K the electric field strength in vacuum is equal to E_0, the vector $\mathbf{E}_0$ forming an angle θ with the normal of the dielectric's surface (Fig. 3.11). Assuming the field to be uniform both inside and outside the dielectric, find:

(a) the flux of the vector $\mathbf{E}$ through a sphere of radius R with centre located at the surface of the dielectric;

(b) the circulation of the vector $\mathbf{D}$ around the closed path Γ of length l (See Fig. 3.11) whose plane is perpendicular to the surface of the dielectric and parallel to the vector $\mathbf{E}_0$.

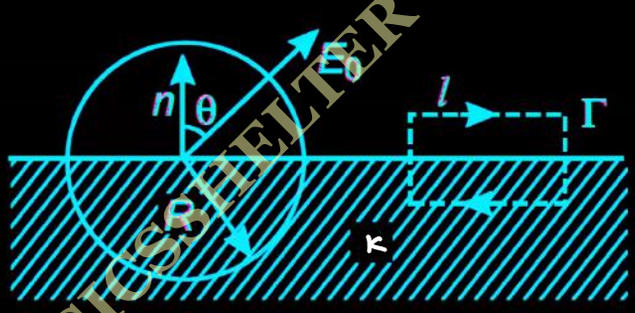


Fig. 3.11

Note: $\vec{D} = \vec{P} + \epsilon_0 \vec{E}$ (Also known as displacement vector)

Solution: Flux can be found by superposition of tangential and normal fields. Now Since tangential field is $E_0 \sin\theta$ which is uniform outside as well as inside, it will create a net zero flux. (Imagine the sphere in uniform tangential field).

Now flux due to Normal field:

$$\phi = E_0 \cos\theta \times \pi R^2 - \frac{E_0 \cos\theta}{K}\pi R^2 = \pi R^2 E_0 \cos\theta\left(1 - \frac{1}{K}\right)$$ Ans

Now $\oint \vec{D}\cdot d\vec{l} = \oint \vec{P}\cdot d\vec{l} + \oint \vec{E}\cdot d\vec{l} = \oint \vec{P}\cdot d\vec{l}$ $(\because \oint \vec{E}\cdot d\vec{l} = 0)$

Now $\vec{P}$ in vacuum is zero, and inside dielectric, $\vec{P} = \chi \epsilon_0 \vec{E}$

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Now $\int \vec{P}\cdot d\vec{l}$ inside $= -(P_T \times l)$ (P_T = tangential polarization)

$= -\chi \epsilon_0 E_0 \sin\theta \times l = -(K-1)\epsilon_0 E_0 l \sin\theta$ Ans

EXAMPLE: IRODOV 3.81

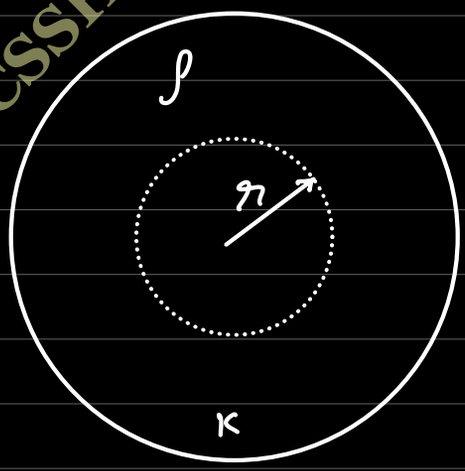
Extraneous charges are uniformly distributed with space density $\rho > 0$ over a ball of radius R made of uniform isotropic dielectric with constant κ. Find:

(a) the magnitude of the electric field strength as a function of distance r from the centre of the ball; draw the approximate plots $E(r)$ and $\phi(r)$; ($\phi \rightarrow$ potential)

(b) the space and surface densities of the bound charges.

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Solution: Consider gaussian surface of radius r. Applying Gauss law for Dielectrics:



$$K \times E \times 4\pi r^2 = \left(\rho \times \frac{4}{3}\pi r^3\right)/\epsilon_0$$

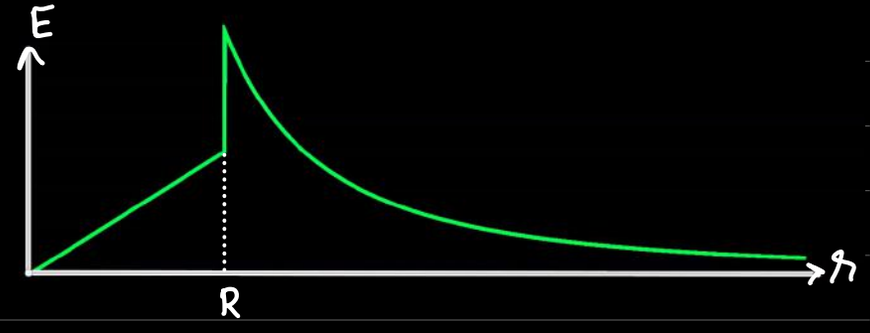
$$\Rightarrow E = \frac{\rho}{3K\epsilon_0} r \quad \text{Ans. for } r < R.$$

Now for $r \geq R$, applying Gauss' law:

$$E \times 4\pi r^2 = \left(\rho \times \frac{4}{3}\pi R^3\right)/\epsilon_0 \Rightarrow E = \frac{\rho R^3}{3\epsilon_0 r^2} \quad \text{Ans.}$$

(Note that there will be sudden jump in field just after crossing the dielectric vacuum boundary).

The adjacent figure shows E vs r graph. Notice the sharp jump at $r = R$.



Now Potential at outside points is just like that of point charges,

Using $E = -\frac{dV}{dr}$, $V = \frac{\rho R^3}{3\epsilon_0 r}$ for $r \geq R$.

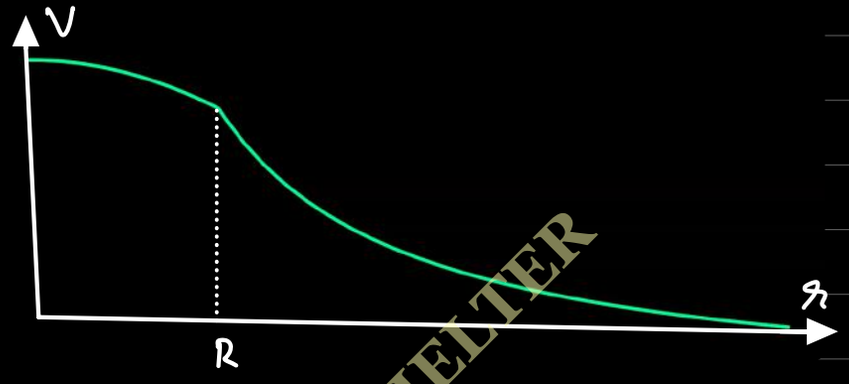
Now for inside points, $-\frac{dV}{dr} = \rho r/3\kappa\epsilon_0 \Rightarrow V = \frac{-\rho r^2}{6\kappa\epsilon_0} + C$

Also @ $r = R$, $V = \frac{\rho R^2}{3\epsilon_0} = \frac{-\rho R^2}{6\kappa\epsilon_0} + C \Rightarrow C = \frac{\rho R^2}{3\epsilon_0}\left[1 + \frac{1}{2\kappa}\right]$

$\Rightarrow V = \frac{\rho}{3\epsilon_0}\left[\frac{R^2 - r^2}{2\kappa} + R^2\right]$

The adjacent figure shows plot of potential Vs r.

Note that at $r = R$, there is a sudden change in slope (because of abrupt change in E).



(b) Now $\vec{\nabla}\cdot\vec{P} = -\rho_b \Rightarrow \chi\epsilon_0\vec{\nabla}\cdot\vec{E} = \rho_b$ $(\because \vec{P} = \chi\epsilon_0\vec{E})$

$\Rightarrow \chi\epsilon_0\left[\frac{dE}{dr} + \frac{2E}{r}\right] = -\rho_b$, $E = \frac{\rho r}{3\epsilon_0\kappa}$

$\Rightarrow -\chi\epsilon_0\left[\frac{\rho}{3\kappa\epsilon_0} + \frac{2\rho}{3\kappa\epsilon_0}\right] = \rho_b = -\frac{(\kappa-1)}{\kappa}\rho$ Ans.

Now at surface, $\sigma_b = \vec{P}\cdot\hat{n} = \chi\epsilon_0\vec{E}\cdot\hat{n} = (\kappa-1)\epsilon_0\frac{\rho R}{3\epsilon_0\kappa}$

$(\because \vec{E} = \rho R/3\epsilon_0\kappa$ near surface). Thus $\sigma_b = \left(\frac{\kappa-1}{\kappa}\right)\frac{\rho R}{3}$ Ans.

EXAMPLE : IRODOV : 3.82 :

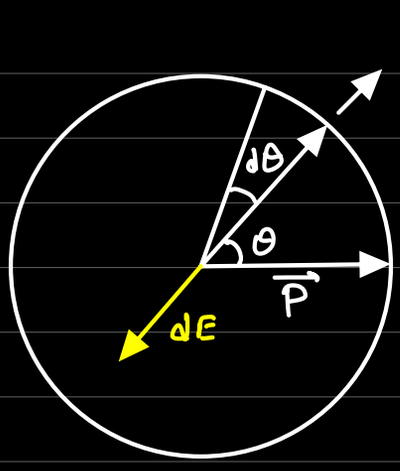
A round dielectric disc of radius R and thickness d is *statically* polarized so that it gains the uniform polarization **P**, with the vector **P** lying in the plane of the disc. Find the strength **E** of the electric field at the centre of the disc if $d \ll R$.

Solution: $\because \vec{P}$ = Constant $\Rightarrow \vec{\nabla}\cdot\vec{P} = 0 \Rightarrow$ There is no volume bound charge and we only need to deal with surface bound charges. Also $\because \vec{P}$ is in plane of disc, flat surfaces of disc won't get any charge ($\because \vec{P}\cdot\hat{n} = 0$ for these surfaces).

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Now charge density on curved surface,

$$\sigma = \vec{P}\cdot\hat{n} = P\cos\theta$$



Now consider a small element of the disc of angular thickness $d\theta$ due to this, at the centre:

$$dE = \frac{1}{4\pi\epsilon_0}\,\frac{P\cos\theta \times R\,d\theta \times d}{R^2},$$

Due to symmetry we only need to integrate its horizontal component. Thus:

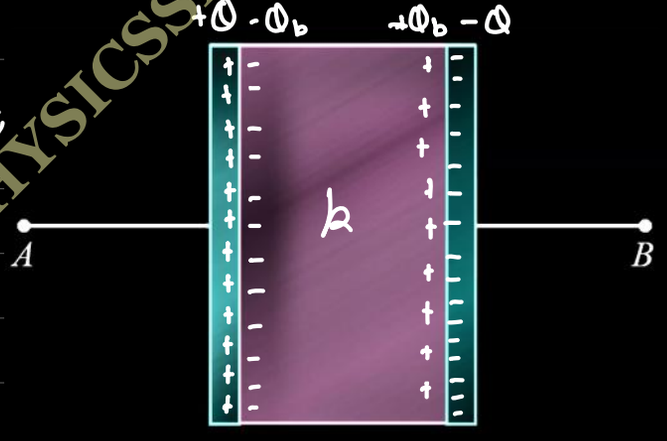
$$E = \frac{Pd}{4\pi\epsilon_0 R}\int_0^{2\pi}\cos^2\theta\,d\theta$$

$$\Rightarrow \vec{E} = \frac{Pd}{4\epsilon_0 R}(-\hat{i}) \quad \text{Ans.}$$

Parallel Plate Capacitor With Dielectric:

If the capacitance of a parallel plate capacitor is C without the dielectric slab, with completely inserted dielectric, the capacitance increases k times i.e

$$\boxed{C' = kC = \frac{\epsilon_0 kA}{d}}$$



This can be shown very easily as follows:

Without dielectric, field in the region between plates, $E = Q/A\epsilon_0$

With dielectric, field: $E' = (Q - Q_b)/A\epsilon_0$

but $Q_b = Q\left(1 - \frac{1}{k}\right)$ ($\because \sigma_b = -\sigma(1 - 1/k)$ recall dielectric conductor boundary).

Thus $E' = Q/kA\epsilon_0 = E/k$. Thus the electric field in the region has reduced by a factor k and therefore potential difference also reduces by the same factor i.e $V' = V/k$.

Now $C = Q/V$

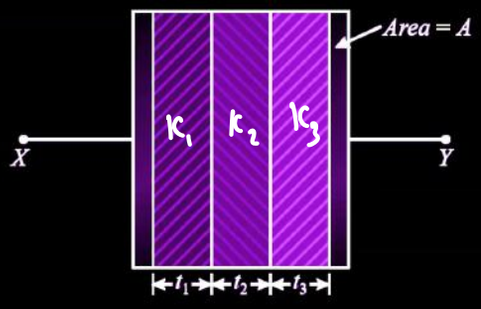
$C' = Q/V' = Q/(V/k) = kQ/V = kC$. QED.

→ If E_0 is field without dielectric with some charge on plates then with dielectric field becomes E_0/k (If charge is kept same).

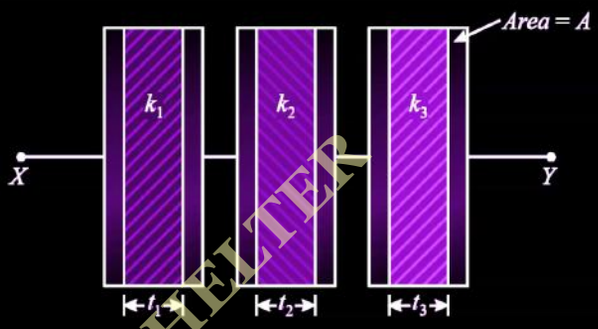
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Parallel Plate Capacitor With Multiple Dielectric Slabs ∥ To Plates:

When multiple slabs are put in a capacitor each with plate area equal to that of the capacitor plates, the system can be broken in to multiple capacitors in series as shown (Imagine inserting thin metal sheets at dielectric boundaries and use the trifurcation technique). Thus here,



$$C_{eff} = \frac{\epsilon_0 A}{t_1/k_1 + t_2/k_2 + t_3/k_3}$$



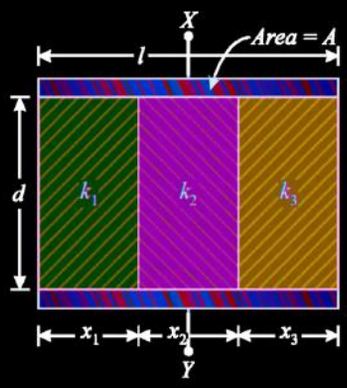
→ If there is an "Air Slab" or Vacuum slab, use the same formula with $k = 1$.

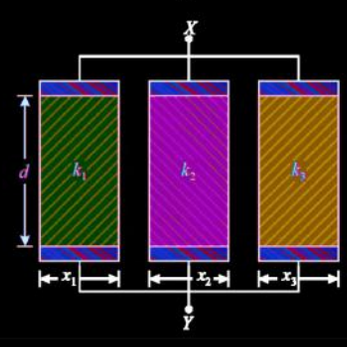
→ It can be readily seen that order of stacking does not matter (for that matter if all slabs were sliced vertically and pieces shuffled, the net capacitance still remains the same).

→ If E_0 is the field without dielectrics for a given charge then with dielectric slab, field simply becomes E_0/k for same total charge. (If charge is scaled up such as due to a battery connection, field gets scaled up by same factor, i.e C'/C)

Parallel Plate Capacitor With Multiple Slabs ⊥ To Plates:

When multiple slabs are put in a parallel plate capacitor with planes ⊥ to the capacitor plates as shown, it can be treated as a parallel combination of multiple capacitor as shown. (It can be readily seen that since potential difference is same across all dielectrics ∴ such capacitors must be in parallel). Here effective capacitance is given as:





$$C_{eff} = \frac{\epsilon_0 b}{d}\left[k_1 x_1 + k_2 x_2 + k_3 x_3\right]$$

Here b is the inward depth of the plates.

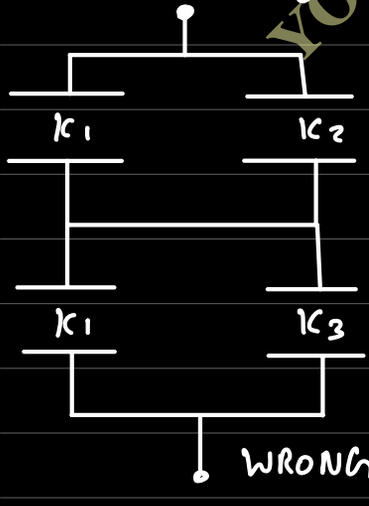
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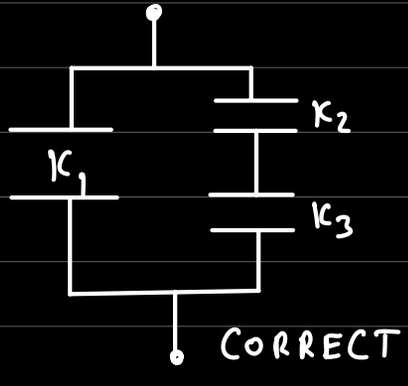
→ Here also it can be seen that order of stacking the slabs does not matter. For that matter if slabs are sliced vertically and shuffled, still the capacitance will remain the same.

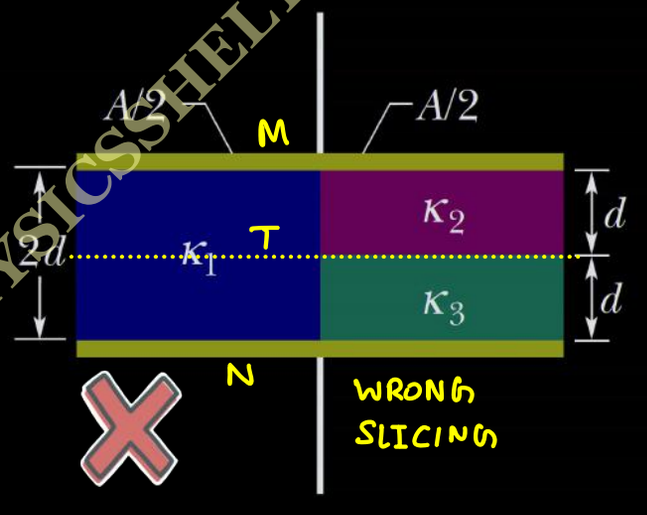
→ Electric field in all plates is equal to V/d where V is the potential difference across plates. A dielectric reduces the field by a factor of K, thus to have same field in all dielectrics, charge densities on conductor must be proportional to dielectric constant [i.e $\sigma_1/k_1 = \sigma_2/k_2 = \sigma_3/k_3$ $\because E = \sigma/k\epsilon_0$ and E is same across all the three dielectrics].

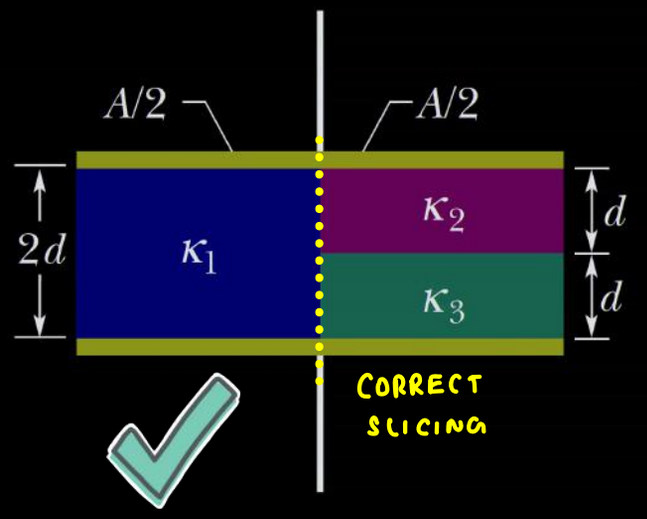
Slicing A Capacitor With Several Dielectrics:

Consider the dielectric arrangement in a capacitor as shown in the figure, Here Total work done in going from plate M to N will be same whether going through K_1 or through K_2 and K_3, ($\because$ M and N are two equipotential surfaces), however work done in going from M to boundary T through K_1 need not be equal to that while passing through K_2, thus boundary T is not equipotential. This makes the partitioning in the first figure invalid. Thus following circuit is a wrong simplification:









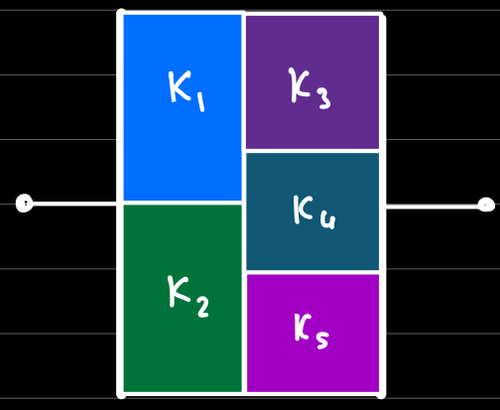
(First partition would have been correct if there were already a metal plate at the yellow boundary).

The second figure shows correct slicing, since M and N are equipotentials we can draw as many vertical planes for partitioning to

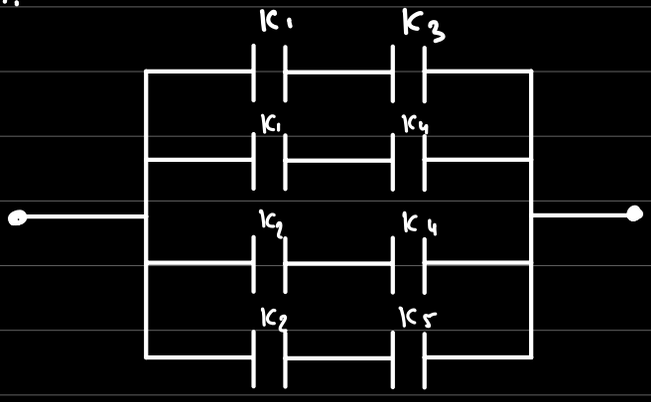
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create parallel capacitor systems.

Think Yourself: Can you draw correct simplification of the shown dielectric arrangement in the parallel plate capacitor.

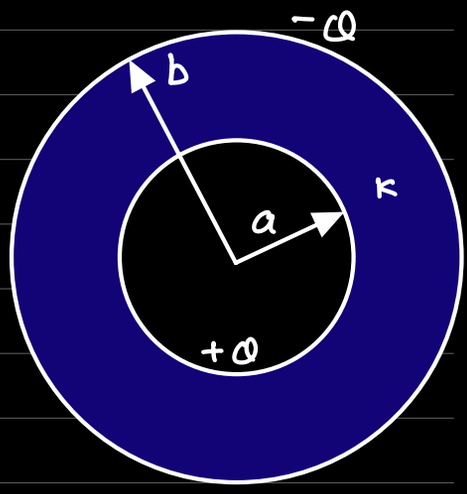


Solution:



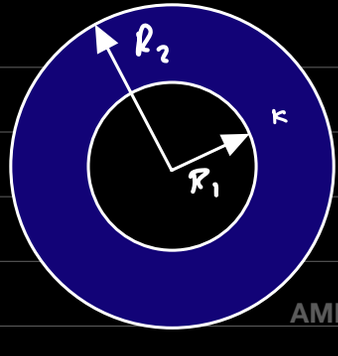
Capacitance Of Concentric Spherical / Cylindrical Shell Capacitors:

If a dielectric completely fills the region between two thin shells, using Gauss law for dielectrics it can be readily seen that, electric field and hence the potential difference for same charge gets scaled down by a factor of K, hence capacitance gets scaled up by the same factor i.e $C' = KC$. If there are several concentric dielectrics, we can imagine a metallic thin shell at each dielectric interface to find the effective capacitance (Introducing metal shell neither changes Q nor V thus C remains same.



Example : IRODOV 3.101:

Find the capacitance of an isolated ball-shaped conductor of radius R_1 surrounded by an adjacent concentric layer of dielectric with constant k and outside radius R_2.



Solution: Recall for isolated conductors, we take the other conductor at infinity. Now we visualise metallic shell just outside dielectric boundary to break the system in to a series

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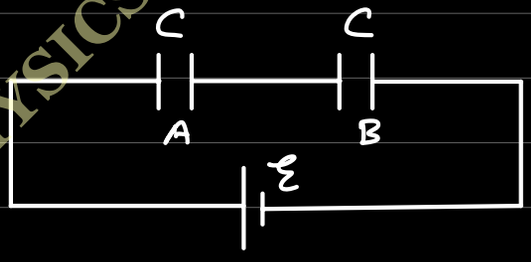
combination of two capacitors $[R_1, R_2, k]$ Series $[R_2, \infty, 1]$
(Say C_1 Ser C_2) where $C_1 = \frac{4\pi\epsilon_0 k R_1 R_2}{R_2 - R_1}$ and $C_2 = 4\pi\epsilon_0 R_2$

$$\Rightarrow C_{eff} = \frac{C_1 C_2}{C_1 + C_2} = 4\pi\epsilon_0 R_2 \left[\left(\frac{kR_1}{R_2 - R_1} \times 1 \right) \Big/ \left(\frac{kR_1}{R_2 - R_1} + 1 \right) \right]$$

$$\Rightarrow C_{eff} = 4\pi\epsilon_0 R_2 \left[\frac{kR_1}{(k-1)R_1 + R_2} \right] \text{ Ans.}$$

EXAMPLE: IRODOV 3.102:

Two parallel-plate air capacitors, each of capacitance C, were connected in series to a battery with emf $\mathcal{E}$. Then one of the capacitors was filled up with uniform dielectric with constant k. How many times did the electric field strength in that capacitor decrease? What amount of charge flows through the battery?



Solution: Initial $C_{eff} = C/2$

$\Rightarrow$ Initial $q_i = C\mathcal{E}/2$

Final

$$C'_{eff} = \frac{C \times kC}{C + kC} = \left(\frac{k}{1+k} \right) C$$

$\Rightarrow$ Final $q' = \left(\frac{k}{k+1} \right) C\mathcal{E}$ $\Rightarrow$ charge flown $= q' - q_i = C\mathcal{E}\left(\frac{k}{k+1} - \frac{1}{2} \right)$

$$\Rightarrow q' = \frac{1}{2}\left(\frac{k-1}{k+1} \right) C\mathcal{E} \text{ Ans.}$$

Now since plate separation is constant, electric field is simply proportional to potential difference which in turn is inversely proportional to capacitance. Now suppose dielectric is inserted in capacitor B, thus

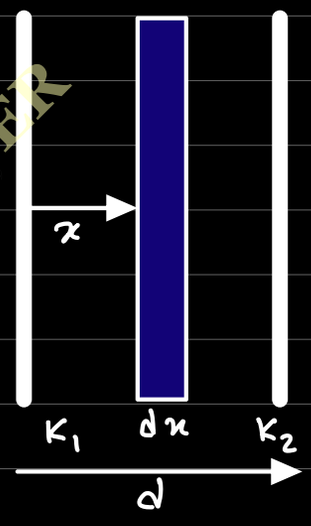
$$V_B' = \left[1/kC \Big/ (1/kC + 1/C) \right] \mathcal{E} = \left(\frac{\mathcal{E}}{1+k} \right)$$

Initial $V_B = \mathcal{E}/2$, thus electric field reduces by a factor V_B'/V_B i.e diminishes by a factor of $2/(1+k)$ Ans.
(There is a misprint in Irodov Key).

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Example: Irodov 3.104:

The gap between the plates of a parallel-plate capacitor is filled with isotropic dielectric whose constants κ varies linearly from κ_1 to κ_2 $(\kappa_2 > \kappa_1)$ in the direction perpendicular to the plates. The area of each plate equals S, the separation between the plates is equal to d.
Find :(a) the capacitance of the capacitor;
(b) the space density of the bound charges as a function of κ if the charge of the capacitor is q and the field $\mathbf{E}$ in it is directed toward the growing κ values.



Solution: We can consider the capacitor to be made of several capacitors of infinitesimal thickness connected in series. Now consider a strip of thickness dx at a location x as shown, for this, $d(1/C) = \dfrac{dx}{\epsilon_0 k S}$ ①

Also $k = k_1 + \left(\dfrac{k_2 - k_1}{d}\right)x$ ②

Putting ② in ① and integrating, we get:

$$1/C = \frac{d}{\epsilon_0 S}\int_0^d \frac{dx}{(k_2-k_1)x + k_1 d} = \frac{d}{\epsilon_0 S}\cdot\frac{1}{(k_2-k_1)}\ln(k_2/k_1)$$

$$\Rightarrow C = \frac{\epsilon_0 (k_2-k_1) S}{d \ln(k_2/k_1)} \text{ Ans.}$$

(b) Now if charge of capacitor is q, local electric field at any layer,

$$E = \left(\frac{q}{S\epsilon_0 k}\right), \quad \text{Now } \rho_b = -\vec{\nabla}\cdot\vec{P}$$

$$\Rightarrow \rho_b = -\vec{\nabla}\cdot(\chi\epsilon_0\vec{E}) = -\frac{d}{dx}\left[(k-1)\epsilon_0\frac{q}{S\epsilon_0 k}\right]$$

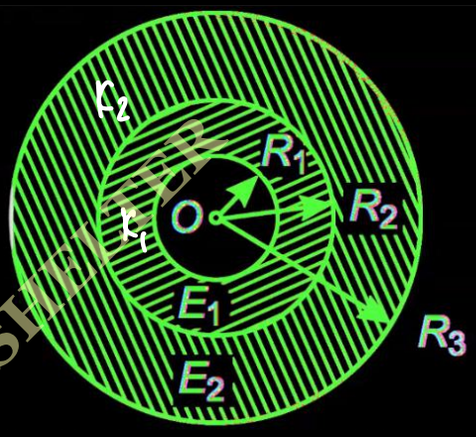
$$\Rightarrow \rho_b = -\frac{q}{S}\frac{d}{dx}\left[1-\frac{1}{k}\right] = -\frac{q}{Sk^2}\left(\frac{dk}{dx}\right) = \frac{-q(k_2-k_1)}{Sk^2 d} \text{ Ans.}$$

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DIELECTRIC BREAKDOWN: If the electric field in a dielectric becomes sufficiently high, electrons start getting ripped off from the bonds and the dielectric starts behaving like a conductor due to presence of free electrons. The field at which this phenomenon starts is called the dielectric breakdown strength (usually specified in units of Volts per metre).

EXAMPLE: IRODOV 4.107:

There is a double-layer cylindrical capacitor whose parameters are shown in Fig. 3.16. The breakdown field strength values for these dielectrics are equal to E_1 and E_2 respectively. What is the breakdown voltage of this capacitor if $\kappa_1 R_1 E_1 < \kappa_2 R_2 E_2$?



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Solution: In a cylindrical capacitor, field declines as $1/r$ so if breakdown happens, it should happen at the innermost layers of respective dielectrics. Let E_0 be the electric field in absence any dielectric at R_1 corresponding to some maximum allowable charge per unit length λ on the capacitor, then

$$E_0 = \frac{\lambda}{2\pi\varepsilon_0 R_1} \quad \text{also} \quad \frac{E_0}{\kappa_1} < E_1 \quad \text{and} \quad E_0 \times \frac{R_1}{R_2} \times \frac{1}{k_2} < E_2$$

$\Rightarrow E_0 < E_1 k_1$ and $E_0 < E_2 k_2 R_2/R_1$, Now $\because E_2 R_2 k_2 > E_1 R_1 k_1$, the first bound is more restricting. Thus:

$$\frac{\lambda}{2\pi\varepsilon_0 R_1} < E_1 k_1, \text{ in limiting case: } \lambda = 2\pi\varepsilon_0 E_1 R_1 k_1$$

Now we just need to find potential difference corresponding to this λ. Thus,

$$V = \frac{\lambda}{2\pi\varepsilon_0}\left[\int_{R_1}^{R_2} \frac{1}{k_1 r}\,dr + \int_{R_2}^{R_3} \frac{1}{k_2 r}\,dr\right] \quad \ldots \quad (\because E = \frac{\lambda}{2\pi\varepsilon_0 k r})$$

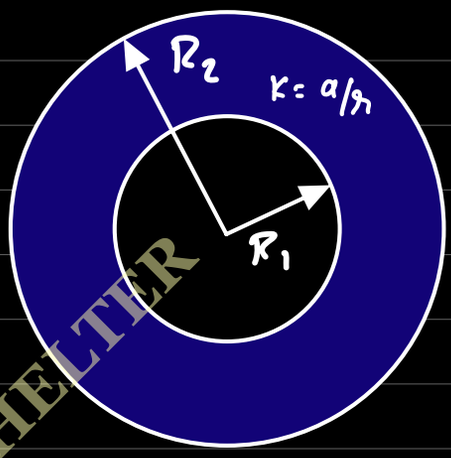
$$\Rightarrow V = \frac{\lambda}{2\pi\varepsilon_0}\left[\frac{1}{k_1}\ln\frac{R_2}{R_1} + \frac{1}{k_2}\ln\left(\frac{R_3}{R_2}\right)\right]$$

$$= E_1 R_1 k_1 \left[\frac{1}{k_1}\ln R_2/R_1 + \frac{1}{k_2}\ln(R_3/R_2)\right] \text{ Ans.}$$

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EXAMPLE: IRODOV 3.105:

Find the capacitance of a spherical capacitor whose electrodes have radii R_1 and $R_2 > R_1$ and which is filled with isotropic dielectric whose constant varies as $\kappa = a/r$, where a is a constant, and r is the distance from the centre of the capacitor.



Solution: Let the charge on capacitor be Q, then field as a function of r:

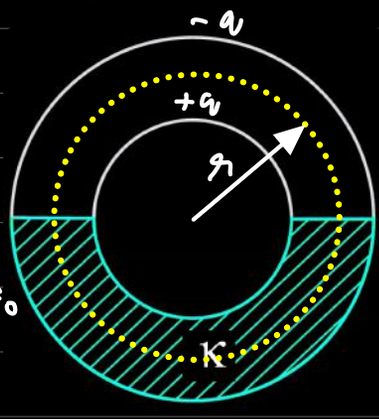
$$E = \frac{Q}{4\pi\epsilon_0 r^2} \times \frac{1}{k} = \frac{Q}{4\pi\epsilon_0 a r}$$

$$\Rightarrow V = \int_{R_1}^{R_2} E\,dr = \frac{Q}{4\pi\epsilon_0 a} \ln(R_2/R_1)$$

$$\Rightarrow C = Q/V = 4\pi\epsilon_0 a / \ln(R_2/R_1) \quad \text{Ans.}$$

EXAMPLE: IRODOV 3.86:

Half the space between two concentric electrodes of a spherical capacitor is filled, as shown in Fig. 3.14, with uniform isotropic dielectric with constant κ. The charge of the capacitor is q. Find the magnitude of the electric field strength between the electrodes as a function of distance r from the curvature centre of the electrodes.



Solution: Imagine a Gaussian surface of radius r as shown in the figure. Now applying Gauss law for dielectrics:

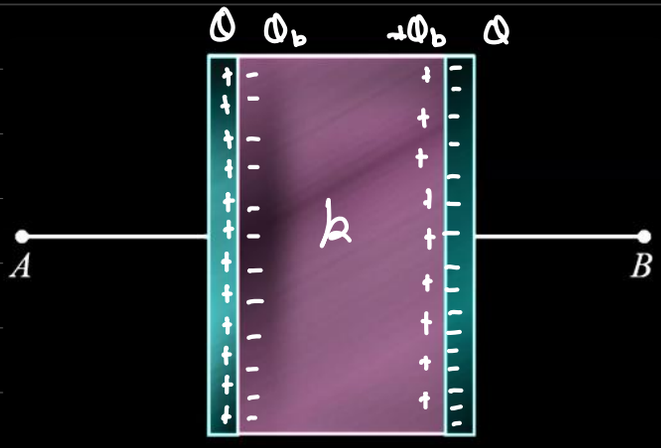
$$\oint k\vec{E}\cdot d\vec{s} = Q_{free_{in}} \Rightarrow k E \times 2\pi r^2 + E \times 2\pi r^2 = q/\epsilon_0$$

$$\Rightarrow E = \frac{1}{2\pi\epsilon_0(\kappa+1)} \cdot \frac{q}{r^2} \quad \text{Ans.}$$

→ Note that two halves of the capacitor will carry unequal charges (1 : κ i.e in ratio of capacitances).

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Energy Density In Dielectrics:



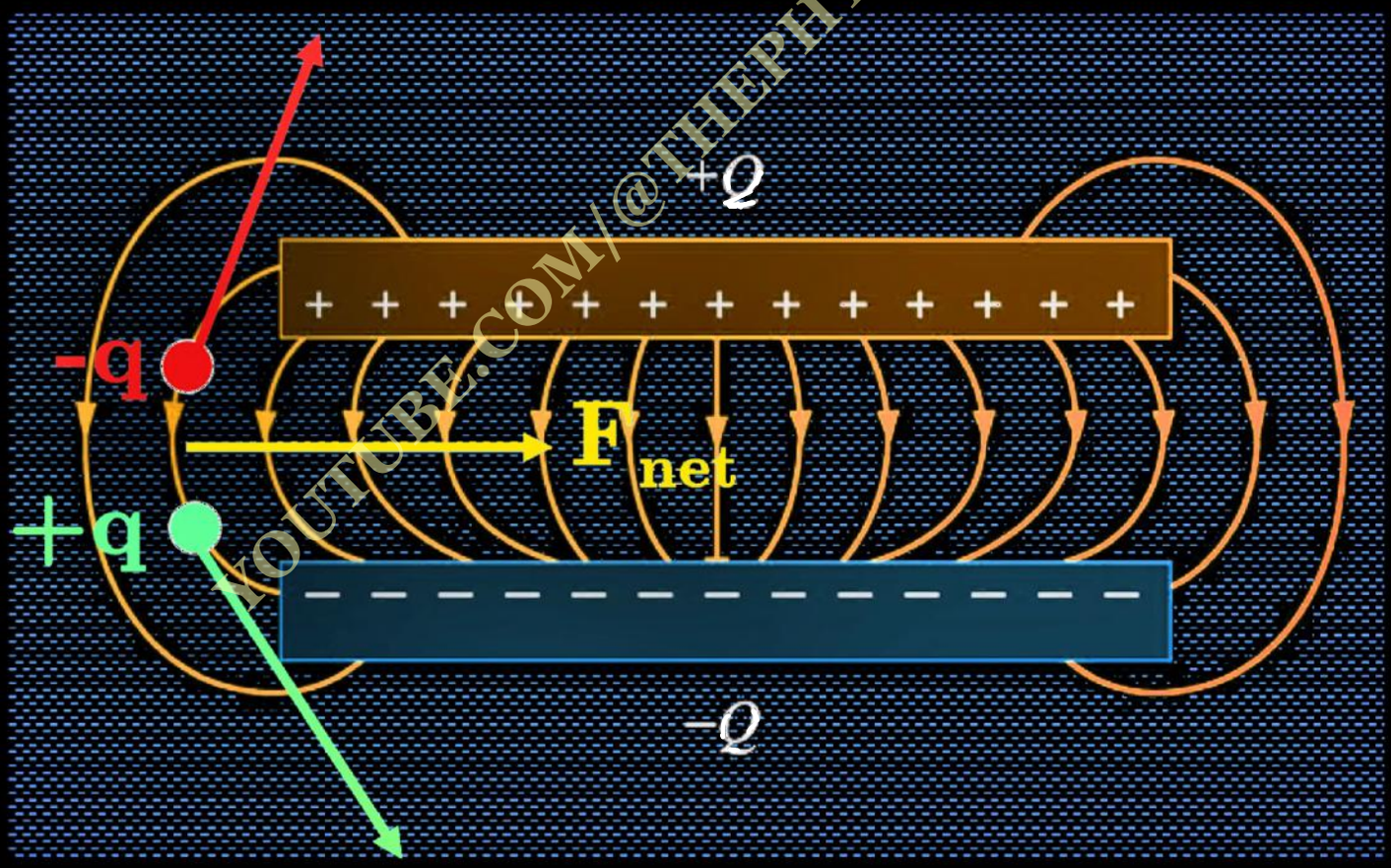
Recall that energy stored in a Capacitor is given as $U = \frac{1}{2} C V^2$, for a capacitor with dielectric we can write this as $U = 1/2 (\epsilon_0 k A/d)(Ed)^2$

$\Rightarrow U = 1/2\ \epsilon_0 k A d E^2$, Now energy per unit volume, $u = U/Ad = 1/2\ \epsilon_0 k E^2$

Thus Energy density in presence of dielectric:

$$u = \frac{1}{2} \epsilon_0 k E^2 = \frac{1}{2} \epsilon E^2$$

Suction Force On A Dielectric Due To Fringing:

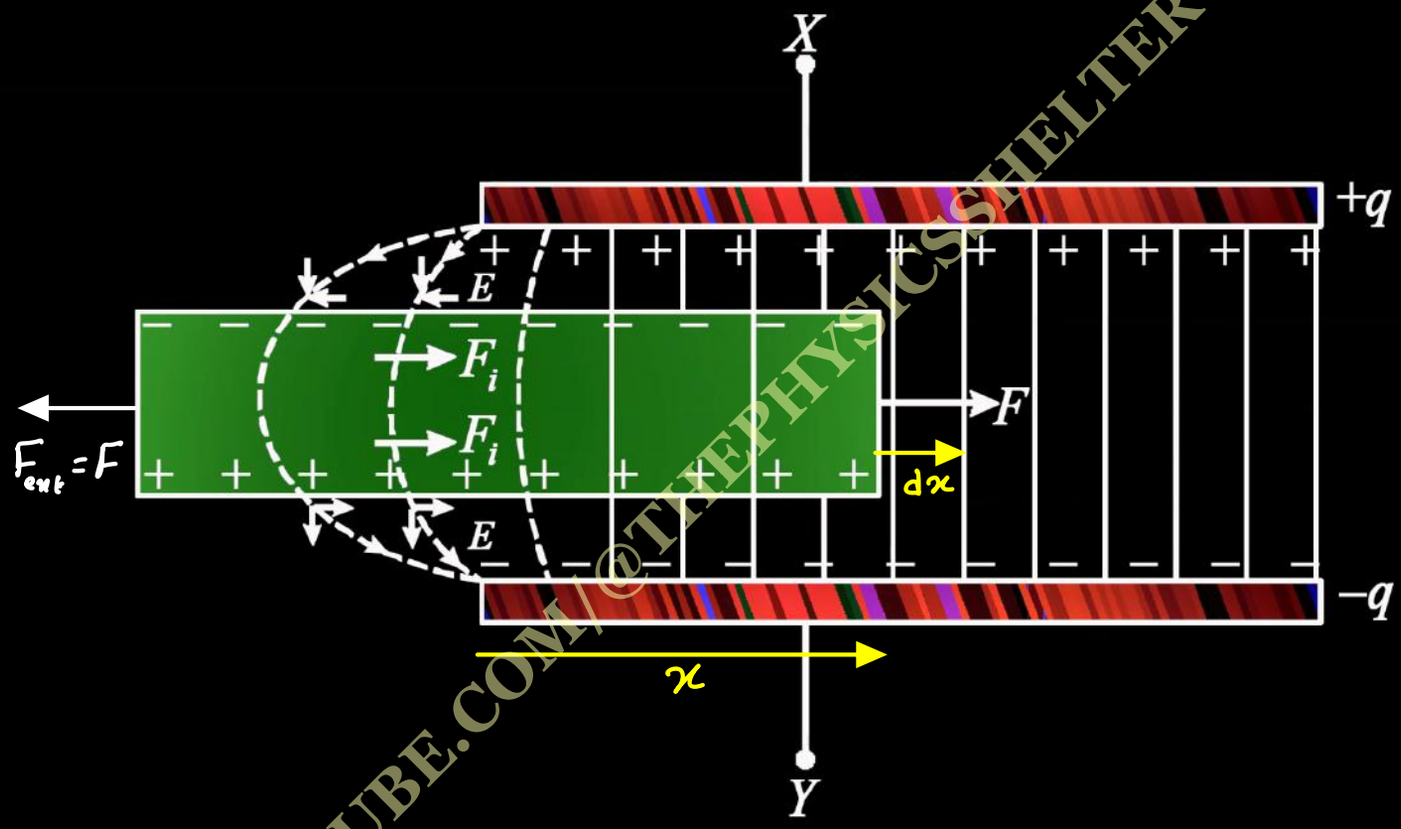


Recall that near the end points of a parallel plate capacitor, field lines are not ⊥ to plates rather they fringe out. Now imagine presence of a dielectric molecule (dipole). One can see from the diagram that this molecule experiences a net inward force and has thus

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a tendency to get sucked inside due to this electrostatic pressure. As a result, if a dielectric slab is put in the Gap, it spontaneously has a tendency to get sucked in to the capacitor. Similarly if the capacitor is immersed in a liquid dielectric, the capacitor pulls the liquid inwards, thus increasing the hydrostatic pressure in the region between the plates. We shall look at these effects mathematically in following sections.

Suction Force On A Dielectric Slab: (Constant Charge)



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Consider a dielectric slab being sucked in to a capacitor as shown in the figure above. If we wish to slowly insert the slab without acceleration, we will need to apply an equal and opposite force using an external agent. Now we know that potential energy of a capacitor is $U = q^2/2C$, let the slab be inserted to a depth x and we further insert it through a distance dx. In this process, the work done by external agent must be equal to change in potential energy stored in capacitor. Thus:

$$-F\,dx = dU \Rightarrow F = -\frac{dU}{dx} = -\frac{d}{dx}\left[\frac{q^2}{2C}\right] = \frac{q^2}{2C^2}\frac{dC}{dx}$$

Thus $\boxed{F = q^2/2C^2\ dC/dx = 1/2\ V^2 dC/dx}$

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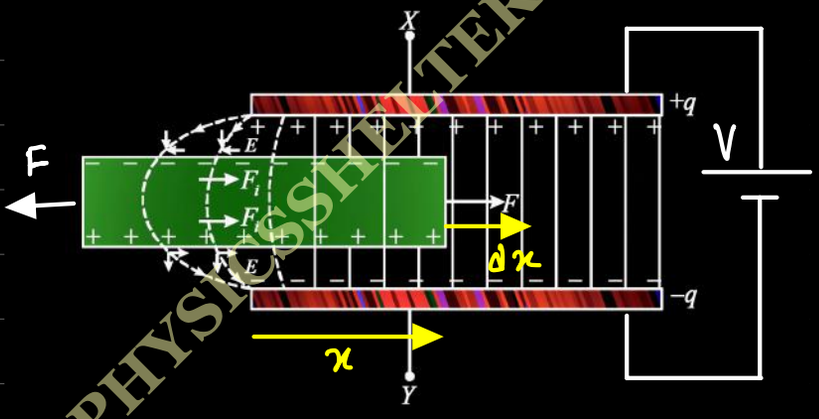
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Here V is the instantaneous Voltage across the capacitor. (If battery is connected, V remains constant, if battery is not connected, q remains constant. Since force only depends on total charge on the capacitor, it does not matter whether this charge exists with battery connected or disconnected, the results for force must be same in both cases, though we will still formally derive for the case with battery connected, just like popular textbooks).

Suction Force On Dielectric Slab (Battery Connected):

Let slab be inserted to a depth x and we further insert a distance dx slowly by applying an external balancing force. Let dq be the charge flown from battery in the process, then, applying work energy theorem

$$-F\,dx + V\,dq = dU$$



Now $U = \frac{1}{2}CV^2 \Rightarrow dU = \frac{1}{2}V^2dC$, $q = CV \Rightarrow dq = V\,dC \Rightarrow V\,dq = V^2dC$

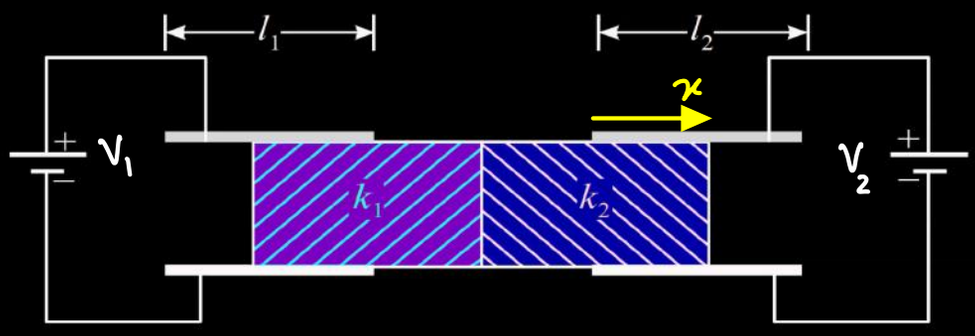
using these substitutions in above equation,

$$-F\,dx + V^2dC = \frac{1}{2}V^2dC \Rightarrow \boxed{F = \frac{1}{2}V^2\frac{dC}{dx}}$$

→ This is the same result that we had claimed in previous section.

→ There is no need to consider any heat loss because there is no heat loss when charges are flowing slowly.

Example: Find V_1/V_2 so that the composite slab does not move. (Capacitors are identical and width of dielectric slab is equal to the width of the plates (inward)).



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Solution: Let the inserted depth in second capacitor be x, then it can be treated as // combination of two capacitors (x length with dielectric and $(l_2 - x)$ length without dielectric)

$\Rightarrow C_2 = \frac{C_{02} k_2 x}{l_2} + \frac{C_{02}(l_2 - x)}{l_2}$ where C_0 is capacitance without dielectric

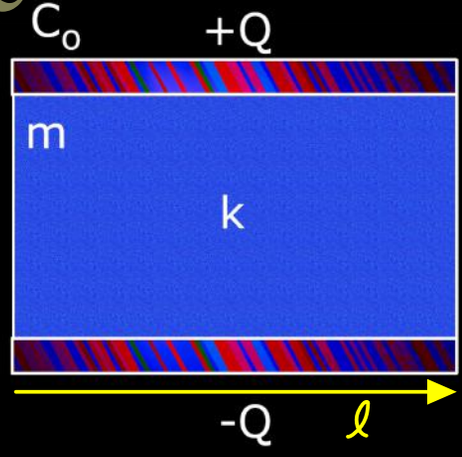
Thus $dC_2/dx = \frac{C_{02}}{l_2}[k_2 - 1]$

Now for equilibrium of composite slab, pulling forces should be identical. Thus:

$$\frac{1}{2} V_1^2 \times \frac{C_{01}}{l_1}[k_1 - 1] = \frac{1}{2} V_2^2 \frac{C_{02}}{l_2}[k_2 - 1]$$

$$\Rightarrow \frac{V_1}{V_2} = \sqrt{\frac{k_2 - 1}{k_1 - 1}} \quad \text{Ans.}$$

EXAMPLE: A capacitor has a capacitance C_0 without dielectric. The capacitor has a charge Q and a length l. Will the motion of the slab be simple harmonic for small displacements. (Plates are fixed, slab can move without friction).



Solution: From previous problem, $dC/dx = \frac{C_0(k-1)}{l}$

Now restoring force $F = \frac{1}{2}\frac{Q^2}{C^2}\frac{dC}{dx}$. Now when slab shifts by a small distance x, $C = \left[\frac{x}{l} C_0 + \left(\frac{l-x}{l}\right) k C_0\right]$

$$\Rightarrow C = C_0\left[k - \frac{x}{l}(k-1)\right] = C_0 k\left[1 - \frac{x}{lk}(k-1)\right]$$

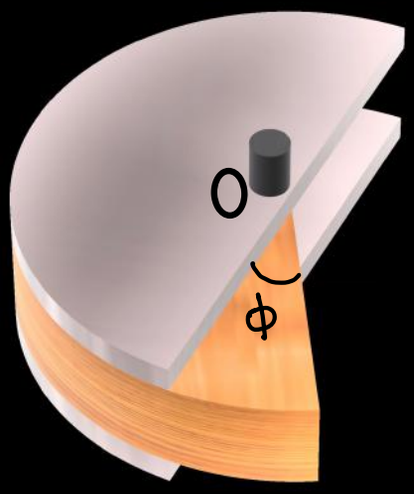
Now $C^{-2} \approx (C_0 k)^{-2}\left[1 + \frac{2x}{l}\left(\frac{k-1}{k}\right)\right]$

Thus $\frac{1}{2} Q^2 C^{-2}(dC/dx)$ is nearly constant for small values of x, thus the motion as a whole is not simple harmonic.

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Example : Irodov 3.146

A capacitor consists of two stationary plates shaped as a semi-circle of radius R and a movable plate made of dielectric with constant k and capable of rotating about an axis O between the stationary plates (See Figure). The thickness of the movable plate is equal to d which is practically the separation between the stationary plates. A potential difference V is applied to the capacitor. Find the magnitude of the moment of forces relative to the axis O acting on the movable plate in the position shown in the figure.



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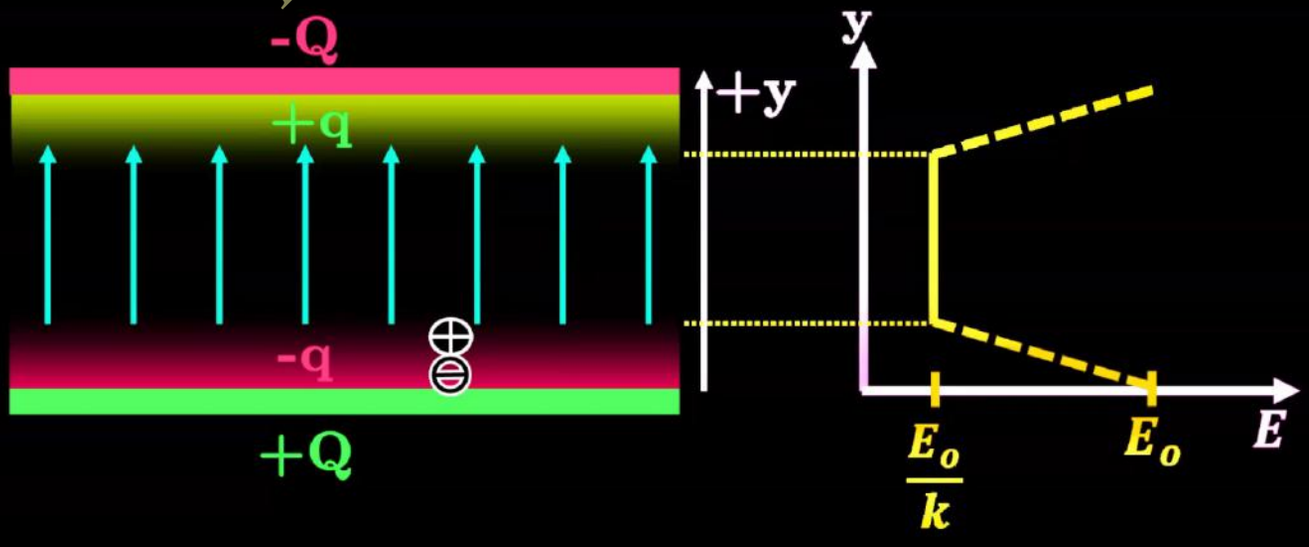
Solution: Just like the problem of finding force on a dielectric with battery connected (which was $\frac{1}{2}V^2 dC/dx$) we can easily prove here that the torque on the dielectric here will be $\frac{1}{2}V^2 dC/d\phi$.
Let an angle ϕ of dielectric be outside the plate region and let us increase it slowly through an angle $d\phi$. Let dC be change in capacitance in the process. Further let the external torque required in the process be τ. Then by work energy theorem:

$$\tau d\phi + V^2 dC = d\left(\frac{1}{2}CV^2\right) = \frac{1}{2}V^2 dC$$

$$\Rightarrow \quad \tau = -\frac{1}{2}V^2\frac{dC}{d\phi} \quad , \text{ Now } \quad C = \frac{\epsilon_0 \pi R^2}{2d}\left(\frac{(\pi-\phi)}{\pi}\right) + \frac{\epsilon_0 \pi R^2}{2d}\left(\frac{\phi}{\pi}\right)$$

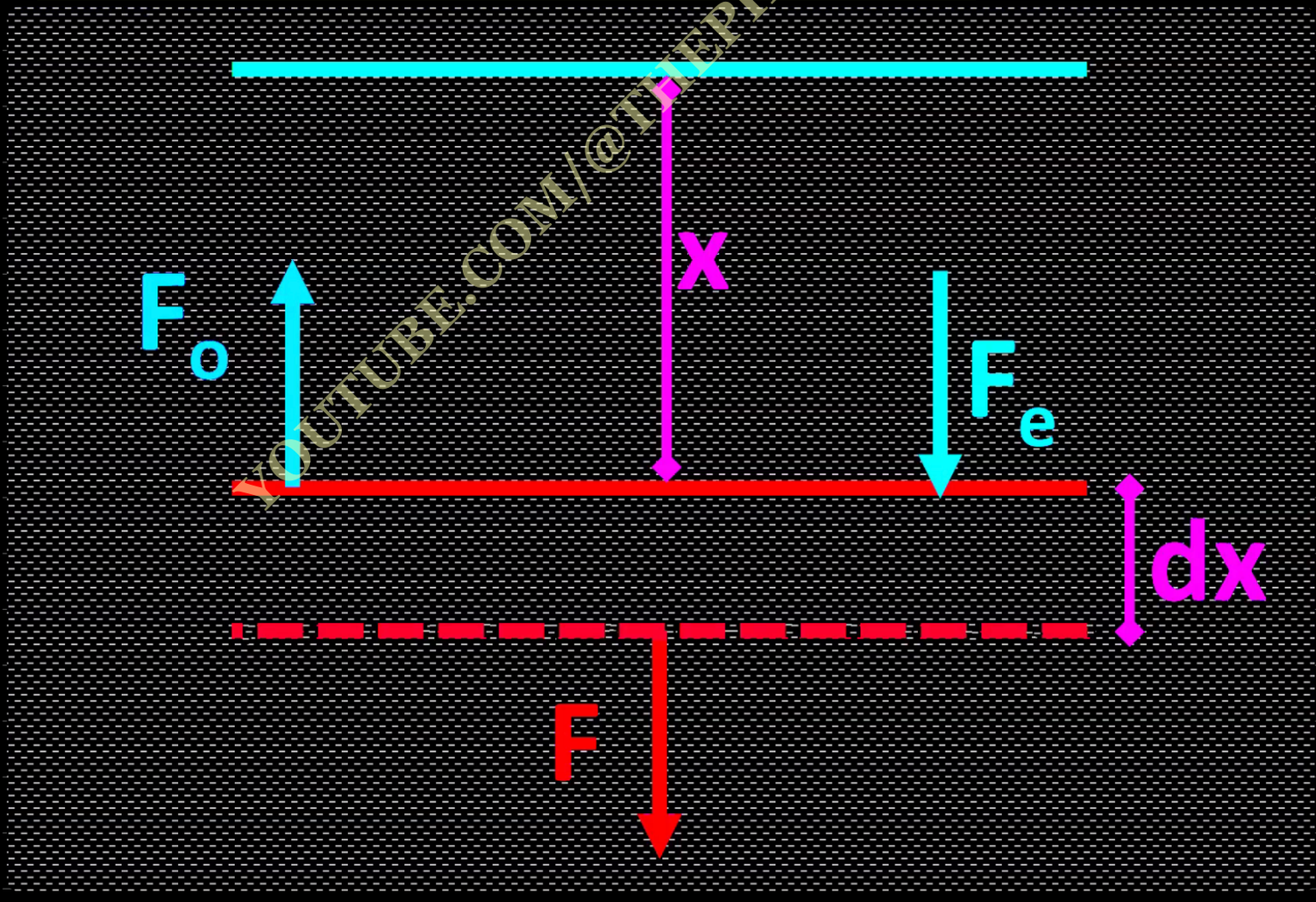
$$\Rightarrow \quad dC/d\phi = \frac{\epsilon_0 \pi R^2}{2d}\left[\frac{-K}{\pi} + \frac{1}{\pi}\right] \Rightarrow \tau = V^2\frac{\epsilon_0 R^2}{4d}[K-1] \text{ Ans.}$$

Field Gradient Near Metal Dielectric Boundary :



So far the dielectric model that we present suggests that there would be a sudden change in electric field as one moves from metal across the dielectric boundary, however, actually the field changes in a finite distance and not abruptly, (There is a gradual decrease in the field magnitude in a few molecules thick layer). Now if we consider a dielectric dipole in this region. Due to gradual change in electric field, the negative charge of dipole near the positive plate experiences a stronger attraction than the repulsion faced by its positive counterpart. Thus this dipole pushes against the positive plate, contributing to excess hydrostatic pressure on metal dielectric boundary. We will try to find out the total excess pressure on the metal dielectric boundary using energy considerations in the next section.

Excess Pressure On Metal Liquid Dielectric Boundary:



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F_0 : Electrostatic force between plates

F_e : Force due to hydrostatic excess pressure

F : External force required to slowly displace the lower plate.

Now: $F_0 = Q^2/2\epsilon_0 A$ (1)

$F = F_0 - F_e$ (2)

Now $U = Q^2/2C = Q^2 x/2\epsilon_0 k A \Rightarrow dU = Q^2 dx/2\epsilon_0 k A = F dx$

$$\Rightarrow F = \frac{Q^2}{2\epsilon_0 k A} = \frac{F_0}{k}$$ Thus: $\boxed{F = F_0/k}$

→ Thus even though electrostatic force between plates is same as that without the dielectric, however the net force (i.e. resultant of Electrostatic and Hydrostatic Forces) has reduced by a factor of k.

Now excess pressure force, $F_e = F_0 - F = F_0 - F_0/k = F_0(1 - 1/k)$

Thus $\boxed{F_e = F_0(1 - 1/k)}$, Now excess pressure $P_e = F_e/A$

$$\Rightarrow P_e = \frac{Q^2}{2A\epsilon_0}(1 - 1/k) \times \frac{1}{A} = \frac{\sigma^2}{2\epsilon_0}(1 - 1/k)$$

Thus:

$$\boxed{P_e = \frac{\sigma^2}{2\epsilon_0}\left(1 - \frac{1}{k}\right)}$$

→ Here excess hydrostatic pressure means the pressure over and above the pressure at the same level in the region outside the capacitor.

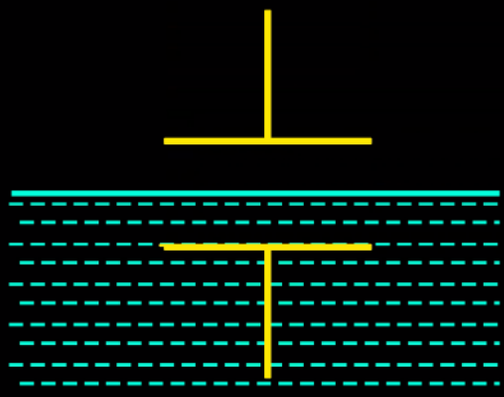
→ Whenever there is a liquid dielectric and metal boundary, there is always an excess hydrostatic pressure given by $\sigma^2/2\epsilon_0(1 - 1/k)$.

→ In case of solid dielectric slabs, the slab gets compressed in direction parallel to the capacitor plate and stretched in direction perpendicular to the plates (if there is clearance), otherwise normal reaction increases if there is no clearance), this phenomenon is called electrostriction.

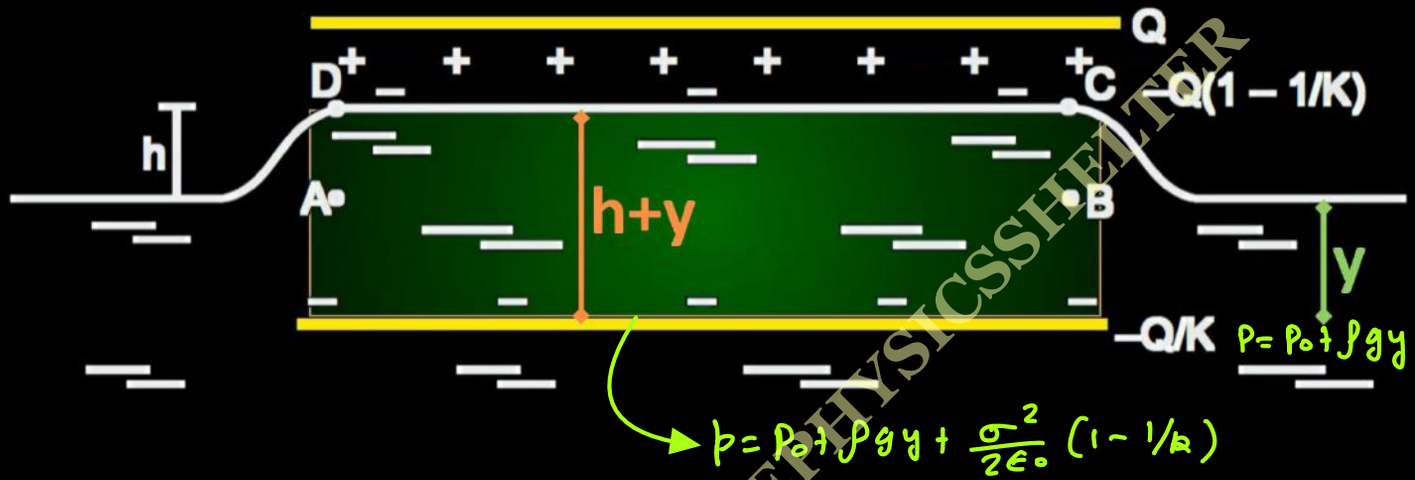
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Example: Irodov: 3.144:

A parallel-plate capacitor is located horizontally so that one of its plates is submerged into liquid while the other is over its surface . The dielectric constant of the liquid is equal to κ and its density is equal to ρ. To what height will the level of the liquid in the capacitor rise after its plates get a charge of surface density σ?



Solution:



Consider the FBD of the dielectric portion shaded in olive green. The net electrostatic force on this is zero, because its net charge is zero, therefore its weight must be balanced by pressure forces. Thus:

$$\left[P_0 + \rho g y + \frac{\sigma^2}{2\epsilon_0}\left(1 - 1/k\right)\right]A - P_0 A = \rho A(h+y)\,g$$

$$\Rightarrow \quad h = \frac{\sigma^2}{2\epsilon_0 \rho g}\left[1 - \frac{1}{k}\right], \quad \text{thus} \quad h = \frac{\sigma^2}{2\epsilon_0 k g \rho}(k-1) \text{ Ans.}$$

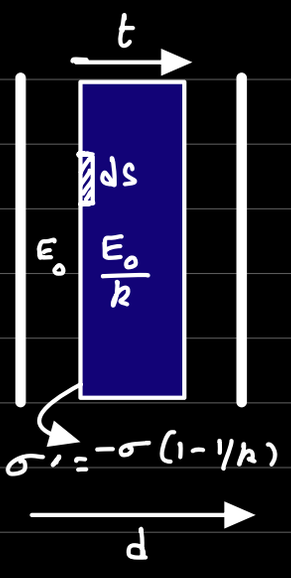
→ A wrong solution to this problem is presented in HC Verma Sir's book (in effect it has assumed same pressure at same level outside and inside).

Example: Pathfinder Check 15:

A dielectric slab of thickness t and relative permittivity k is placed between two fixed parallel metal plates. Faces of the slab and the plates are parallel and distance between the plates is d. Neglecting end effects, find the minimum voltage applied between the plates sufficient to rupture the slab. Breaking stress of the material of the slab is σ_b.

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Solution: We shall try to estimate stress on surface of a dielectric. We consider a patch of area dS on surface of dielectric. Let E_1 be the field contribution to this patch in its vicinity and E_2 be the field due to rest of the universe.



$\Rightarrow$

$$E_1 + E_2 = E_0 \quad (1)$$

$$-E_1 + E_2 = E_0/k \quad (2)$$

$$\Rightarrow E_1 = E_0\left(\frac{k-1}{2k}\right) \qquad E_2 = E_0\left(\frac{k+1}{2k}\right)$$

Now force on the local patch, $dF = E_2 \times \sigma' dS = \frac{\sigma}{\epsilon_0}\left[\frac{k+1}{2k}\right]\cdot\left(-\sigma\left(1-\frac{1}{k}\right)\right]dS$

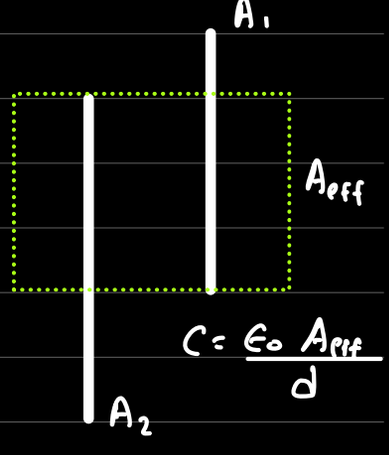
$\Rightarrow$ Pressure: $p = \frac{dF}{dS} = \frac{\sigma^2}{2\epsilon_0 k^2}\left[k^2-1\right] = \sigma_b$ (Breaking stress).

Now $V = \frac{\sigma}{\epsilon_0}[d-t] + \frac{\sigma}{k\epsilon_0}[t] = [kd-(k-1)t]\sqrt{\frac{2\sigma_b}{\epsilon_0(k^2-1)}}$

→ Note here stress created is $\boxed{\frac{\sigma^2}{2\epsilon_0}\left[1-\frac{1}{k^2}\right]}$

Capacitance Of Parallel Plate Capacitor With Unequal Or Skew Areas

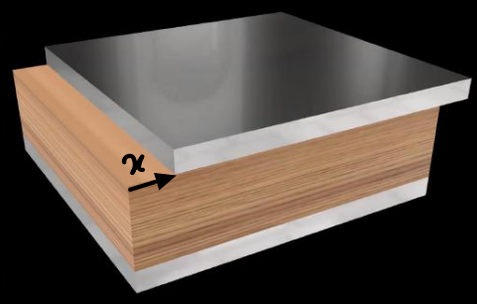
When we have plates of unequal areas or skew areas as shown in the figure (small separation), the effective plate area can be considered only as the parts of areas that are directly facing each other (for purpose of solving problems). Though practically there will also be some fringing effects, which we will simply ignore at JEE level to avoid mathematical complexities (in most of the situations).



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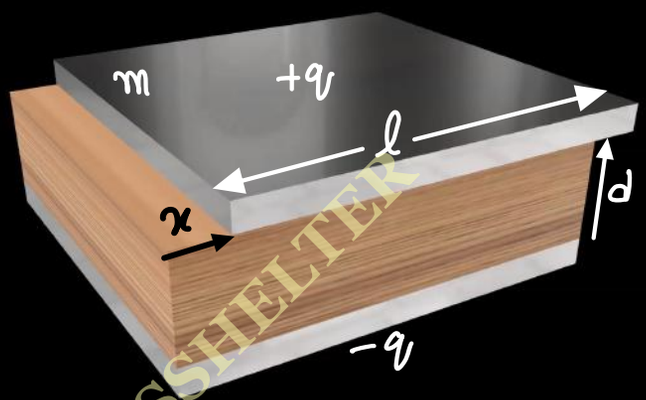
Example: Pathfinder Check: 26:

A parallel plate capacitor completely filled with a dielectric sheet of thickness d and relative permittivity k is placed on a horizontal tabletop. The lower plate is glued with the tabletop and the dielectric slab is glued with the lower plate. The plates are square of edge length l and mass of the upper plate is m. The plates are given charges $+q$ and $-q$, then the upper plate is shifted horizontally by a small distance x along one of its edges and set free. Find period of oscillation of this plate. Thickness d of the dielectric is much smaller than the edge length l.



Solution: When the plate is displaced by some distance x, its potential energy increases because its total charge is still the same however effective capacitance has reduced.

Now restoring force $F = -\frac{\partial U}{\partial x}$.



$$U = \frac{q^2 d}{2\epsilon_0 k l^2 (l-x)/l} = \frac{q^2 d}{2\epsilon_0 l^2 k}\left[1-\frac{x}{l}\right]^{-1} \approx \frac{q^2 d}{2\epsilon_0 l^2 k}\left[1+\frac{x}{l}\right]$$

$$\Rightarrow \text{Restoring Force } F = -\frac{dU}{dx} = \frac{-q^2 d}{2\epsilon_0 k l^3} \Rightarrow a = \frac{F}{m} = \frac{q^2 d}{2\epsilon_0 k l^3 m}$$

Now $T = \sqrt{\frac{2x}{a}} \times 4$ (∵ Four symmetric trips) $(\because \sqrt{2h/g})$.

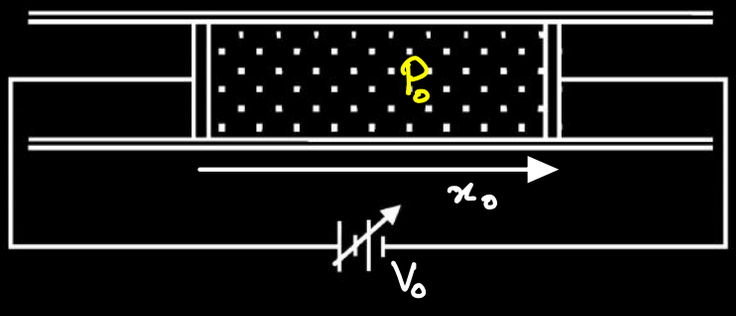
$$\Rightarrow T = \frac{8l}{q}\sqrt{\frac{\epsilon_0 k m l x}{d}} \quad \text{Ans.}$$

Example: Pathfinder: Check: 16

An ideal gas is trapped between two metallic pistons A and B inside a thin glass tube. The pistons are connected with the terminals of a variable voltage battery. Relative permittivity of the gas is ε_r. The pistons stay in equilibrium at separation x_0 between them when battery voltage is V_0.

(a) Is the equilibrium stable, unstable or neutral?

(b) How will the distance x between the pistons change, if battery voltage is slowly increased?



Solution: $C = \frac{\epsilon_0 k A}{x_0}$ Now Electrostatic force $= \frac{Q^2}{2A\epsilon_0} = \left(\frac{\epsilon A}{x_0} V\right)^2 \cdot \frac{1}{2A\epsilon_0}$

$$\Rightarrow F_E = \frac{V^2 A \epsilon_0 k^2}{2x_0^2} = P_0 A \quad (P_0 \text{ is gas pressure in equilibrium}).$$

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Now if separation is decreased by small amount, we shall see the effect on resultant of electrostatic and gas pressure forces. If resultant is outwards, equilibrium is stable (restoring) otherwise it will be unstable.

Now change in electrostatic force : $dF_E = \frac{V^2 A \epsilon_0 k^2}{2} \times \frac{2}{x_0^3} x$

($\because$ Differentiating F_{0E} wrt x_0). $\Rightarrow$ $dF_E = F_{0E} \times \frac{2x}{x_0}$ ①

Now increase in gas spring force $dF_g = \frac{BA}{x_0}(x)$

($\because$ Recall air spring constant) Now $B = \gamma P_0$ thus

$dF_g = \frac{\gamma P_0 x}{x_0} \times A = F_{0_E} \cdot \frac{\gamma x}{x_0}$ ② ($\because$ F_{0_E} is also $P_0 A$ due to equilibrium)

But $\gamma < 2$ thus increase in Gas spring force is less than the increase in electrostatic attraction $\therefore$ equilibrium must be unstable.

// **VIDEO ERRATA:**

Now when separation is decreased by x (not necessarily small),

$F_E = F_{0E} \times \frac{x_0^2}{(x_0 - x)^2} \times \frac{V^2}{V_0^2}$ (See the expression for F_{0E}, we just made adjustments for increased voltage and reduced separation)

Now during adiabatic compression, PV^γ = Constant.

$\Rightarrow$ $P_0 x_0^\gamma = P(x_0 - x)^\gamma \Rightarrow P = \left(\frac{x_0}{x_0 - x}\right)^\gamma \Rightarrow F_g = F_{0g} \times \left(\frac{x_0}{x_0 - x}\right)^\gamma$

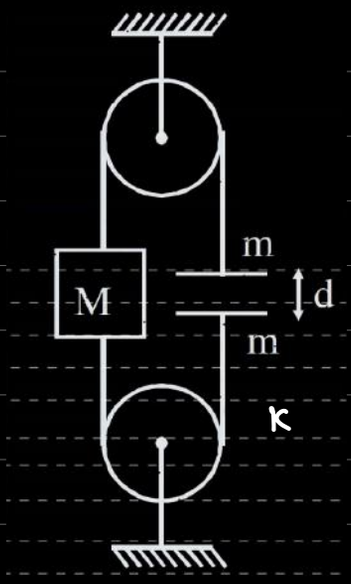
Now $F_E / F_g = \frac{V^2}{V_0^2} \times \left(\frac{x_0}{x_0 - x}\right)^{2-\gamma}$ But $V > V_0$ and $\gamma < 2$ (and $F_{0E} = F_{0g}$)

Thus F_E is always more than F_g, thus equilibrium, thus upon increasing the voltage equilibrium is never obtained and system must collapse to zero volume.

— END ERRATA //

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Example: The shown system is immersed in a liquid of dielectric constant K and negligible density. Find (a) The ratio m/M for equilibrium.
(b) If plate Area is A find the minimum voltage that must be applied across capacitor for equilibrium to be possible.



Solution: For equilibrium weight on LHS and RHS must be same, thus $M = 2m$ Ans (a).

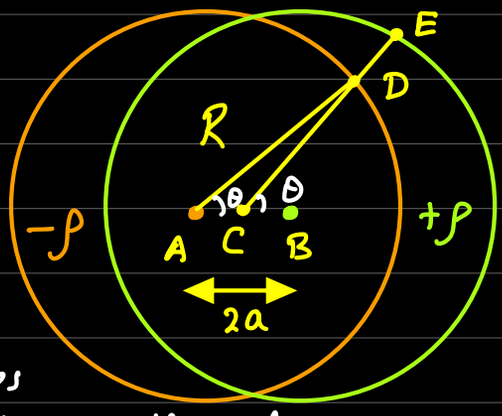
(b) For equilibrium to be possible, in the limiting case, tension in the lower string should be just greater than zero (or equal to zero). In that case mg must be balanced by the resultant of excess pressure force and electrostatic force.

Thus $\frac{M}{2}g = \frac{Q^2}{2hA\epsilon_0} = \left[\left(\frac{\epsilon_0 kA}{d}\right)V\right]^2 \cdot \frac{1}{2hA\epsilon_0}$

$\Rightarrow V = d\sqrt{\frac{Mg}{\epsilon_0 kA}} = \sqrt{\frac{Mgd}{C}}$

Example: A sphere is statically polarized uniformly with polarization $\vec{P}$, find the field everywhere in space because of this. Given radius of sphere is R.

Solution: $\because \vec{P}$ is uniform therefore $\vec{\nabla}\cdot\vec{P} = 0$ thus there is no volume bound charge. Now surface charge at a radius making θ angle with $\vec{P}$, $\sigma_b = \vec{P}\cdot\hat{n} = P\cos\theta$.



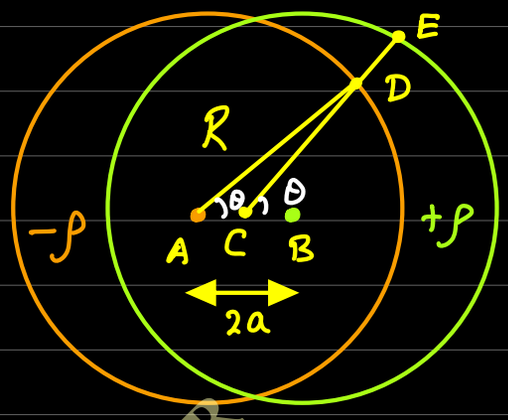
Thus surface charge density is $\sigma_b = P\cos\theta$
Now recall that this can be modeled as a separation of two uniformly charged spheres when $\sigma = \sigma_0\cos\theta$, then field inside was $\sigma_0/3\epsilon_0$ thus here, inside the sphere field is $P/3\epsilon_0$ Ans. Now for outside points, it behaves as a point dipole with dipole moment:

$\vec{p} = \rho \times \frac{4}{3}\pi R^3 \times 2a$ where $2\rho a = \sigma_0 = P$, thus dipole moment is $\vec{p} = \frac{4}{3}\pi R^3 P^3$ (We could also write is directly using definition of polarization vector)

Now for outside the sphere, $\vec{E} = \frac{2kp\cos\theta}{r^3}\hat{r} + \frac{kp\sin\theta}{r^3}\hat{\theta}$ Ans.

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